

Two circles; the central angle and its arc

How two circles can meet, and the angle that measures an arc exactly.

LESSON 55

1 × 40 MIN

GEOMETRY 8, TEMA 56

STAGE 9 · 7.1

LESSON CLOCK

6'

12'

14'

4'

4'

Move two paper circles together	6 min
Positions of two circles	12 min
Central angle and arc	14 min
Arc length again	4 min
Homework	4 min

LEARNING OBJECTIVES

- Classify the position of two circles from the distance between their centres.
- Define the central angle and the degree measure of an arc.
- Find the arc length and the sector area from the central angle.
- Use the fact that equal arcs subtend equal central angles.

TEXTBOOK

National	Tema 56, pp. 132–134
Cambridge	Learner’s Book pp. 142–148
Workbook	Workbook 7.1

01

Explanation

Two circles

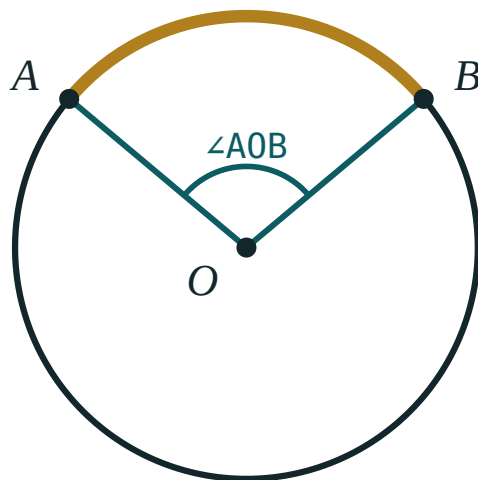
Let d be the distance between the centres and R, r the radii, with $R \geq r$:

CONDITION	POSITION	COMMON POINTS
$d > R + r$	outside each other	0
$d = R + r$	externally tangent	1
$R - r < d < R + r$	intersecting	2
$d = R - r$	internally tangent	1
$d < R - r$	one inside the other	0
$d = 0$	concentric	0

Note the pattern: the sum $R + r$ controls the outside cases, the difference $R - r$ the inside ones. Two circles touch exactly when d hits one of those two values.

The central angle measures the arc

An angle with its vertex **at the centre** is a **central angle**. It cuts off two arcs; the smaller is the **minor** arc, the larger the **major** arc.



a central angle equals the arc it cuts

A central angle and the arc it cuts off have the same degree measure.

DEGREE MEASURE OF AN ARC

The degree measure of an arc equals the central angle that subtends it.

$$\text{arc } AB = \angle AOB$$

A whole circle is 360° , a semicircle 180° , a quarter 90° .

Degrees and length are different things. A 60° arc is always one sixth of its circle, but its **length** depends on the radius:

$$\ell = \frac{\alpha}{360^\circ} \cdot 2\pi r \mid S_{\text{sector}} = \frac{\alpha}{360^\circ} \cdot \pi r^2$$

EQUAL ARCS, ONE CIRCLE

Equal central angles give equal arcs **in the same circle** (or in circles of equal radius). A 60° arc of a large circle is far longer than a 60° arc of a small one.

02 Worked examples

EXAMPLE 1 Two circles have radii 8 and 3, and their centres are 11 apart. Describe their position.

① $R + r = 11$

② $d = 11 = R + r$

Exactly the sum.

③ They touch externally at one point.

The point of contact lies on the line of centres.

ANSWER

Externally tangent

EXAMPLE 2 A central angle of 72° is drawn in a circle of radius 10. Find the arc it cuts off and the sector area.

① $\frac{72}{360} = \frac{1}{5}$

One fifth of the circle.

② $\ell = \frac{1}{5} \cdot 20\pi = 4\pi \approx 12.6$

③ $S = \frac{1}{5} \cdot 100\pi = 20\pi \approx 62.8$

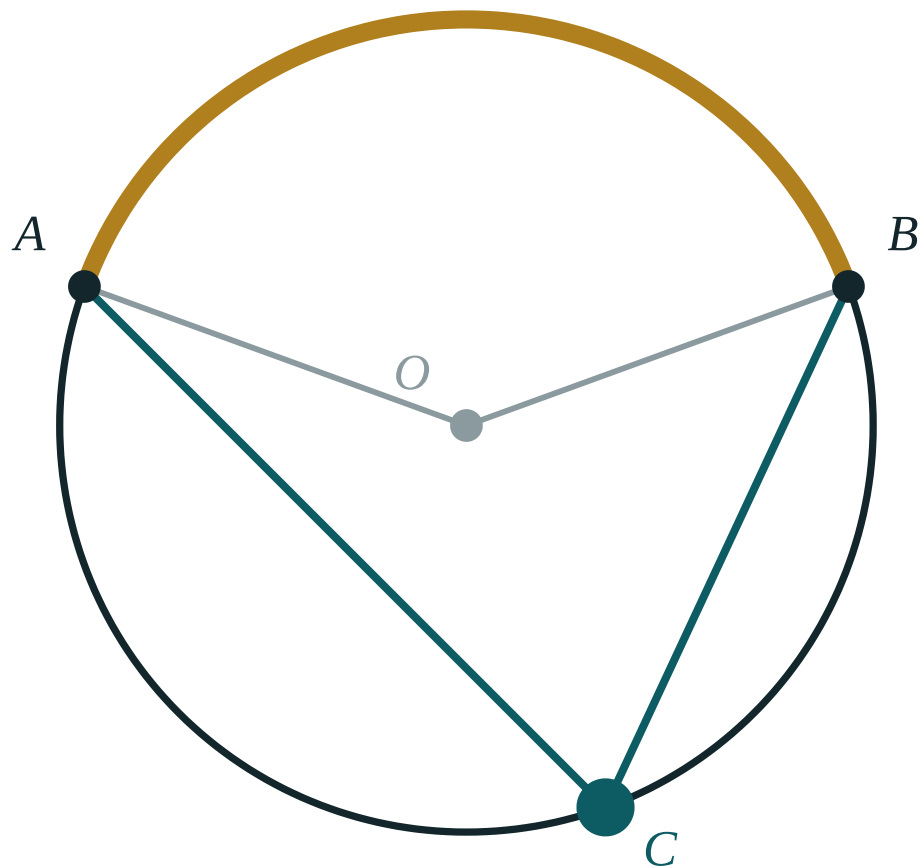
ANSWER

$\ell = 4\pi, S = 20\pi$

03 Interactive model

Drag the points and compare the central angle with the arc it cuts off.

The central angle



central $\angle AOB$ **140°**

inscribed $\angle ACB$ **70°**

ratio **2**

arc AB **140°**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Central angle	Markaziy burchak	Центральный угол
Arc	Yoy	Дуга
Degree measure of an arc	Yoynig gradus o'lchovi	Градусная мера дуги
Concentric circles	Konsentrik aylanalar	Концентрические окружности
Internally tangent	Ichdan urinuvchi	Касающиеся внутренним образом
Externally tangent	Tashqaridan urinuvchi	Касающиеся внешним образом
Intersecting circles	Kesishuvchi aylanalar	Пересекающиеся окружности
Line of centres	Markazlar chizig'i	Линия центров
Major / minor arc	Katta / kichik yoy	Большая / меньшая дуга

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

Circles of radii 5 and 3 with $d = 8$ are:

Circles of radii 9 and 4 with $d = 5$ are:

A central angle of 90° cuts off an arc of:

A 60° arc of a circle of radius 6 has length:

Concentric circles have:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. $R = 6, r = 2, d = 10$. Describe the position.

ANSWER Separate.

2. $R = 6, r = 2, d = 8$. Describe the position.

ANSWER Externally tangent.

3. $R = 6, r = 2, d = 4$. Describe the position.

ANSWER Internally tangent.

4. $R = 6, r = 2, d = 5$. Describe the position.

ANSWER Intersecting.

5. Find the arc measure of a 45° central angle.

ANSWER 45°

6. Find the length of a 90° arc in a circle of radius 8.

ANSWER 4π

7. Find the sector area of a 180° sector of radius 4.

ANSWER 8π

Medium · the standard question

7 PROBLEMS

1. $R = 10, r = 4, d = 14$. Describe the position and the number of common points.

ANSWER Externally tangent, 1 point.

2. $R = 10, r = 4, d = 3$. Describe the position.

ANSWER One inside the other, no common points.

3. Find the length of a 120° arc in a circle of radius 9.

ANSWER 6π

4. Find the sector area of a 120° sector of radius 9.

ANSWER 27π

5. An arc of 40° has length 4π . Find the radius.

ANSWER $r = 18$

6. A sector of 60° has area 6π . Find the radius.

ANSWER $r = 6$

7. Two circles of radius 5 have centres 6 apart. How many common points?

ANSWER 2 — intersecting

Hard · stretch and reason

7 PROBLEMS

1. Two circles of radii 7 and 3 touch internally. Find the distance between the centres, and again if they touch externally.

ANSWER 4 internally, 10 externally

2. A circle of radius 12 has an arc of length 8π . Find its central angle.

ANSWER 120°

3. A sector of radius 10 has area 15π . Find its angle and arc length.

ANSWER 54° , arc 3π

4. Two circles of radius 6 intersect, and their centres are 6 apart. Find the angle subtended at each centre by the common chord.

ANSWER 120° — the triangle of two radii and the line of centres is equilateral, twice over

5. For what values of d do circles of radii 8 and 5 intersect at two points?

ANSWER $3 < d < 13$

6. A running track is a circle of radius 50 m. How long is the arc of a 30° turn?

ANSWER $\frac{1}{12} \cdot 100\pi \approx 26.2$ m

7. Explain why two arcs of the same degree measure can have different lengths.

ANSWER Degree measure records what fraction of its own circle an arc is; length also depends on how big that circle is, through the factor $2\pi r$.

07 Homework

Homework — 5 problems

Geometry 8, Tema 56, pp. 132–134. Compare d with $R + r$ and $R - r$ every time.

- $R = 9, r = 4, d = 13$. Describe the position.
- $R = 9, r = 4, d = 5$. Describe the position.
- Find the length of a 150° arc in a circle of radius 12.

4. *Find the sector area of a 45° sector of radius 8.*
5. *An arc of 60° has length 5π . Find the radius.*